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Sentiment-Driven Predictive Models for Nifty 50 Stock Market Fluctuations: A Time Series Analysis Perspective

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***Abstract*—This paper conducts a thorough semantic analysis of NIFTY-50 stock data, aiming to unravel the intricate dynamics of the Indian stock market. Employing a versatile blend of analytical and machine learning techniques, we delve into the historical performance of individual constituents of the NIFTY-50, seeking valuable insights into trends, patterns, and behavior. Special emphasis is placed on sentiment analysis, recognizing the crucial role of news, external factors, and market sentiment in influencing stock price movements. The utilization of time series forecasting techniques is a key aspect of our investigation, as these methods prove essential in financial analysis for accurate stock price predictions. This study underscores the significance of such techniques and their applicability to investors, financial analysts, and market participants. In essence, this paper serves as a valuable resource for anyone interested in navigating the complexities of the stock market. Our goal is to facilitate well-informed investment decisions and provide a foundation for further exploration in stock market analysis by incorporating a diverse range of techniques and approaches. .**

***Index Terms*—Stock Price Prediction, Forecasting, Time Series Analysis, Big-Data Analytics, Machine Learning.**

I. Introduction

The term ”stock” designates equity or ownership in a company or other organization. Shares are typically awarded in lieu of these stocks. A stock market is characterized as a venue for the trading of bonds and derivatives alongside shares or stocks of publicly traded companies [1]. It serves as a marketplace and platform that links buyers and sellers. There are two main stock exchanges in India. The National Stock Exchange and the Bombay Stock Exchange (BSE) [2]. The regulation of the stock market in India is done by SEBI or the Securities Exchange Board of India. It makes sure that the securities market in India works efficiently and transparently. It protects the interests of all the participants and ensures that no one gets any undue advantages [3]. The most ancient exchange for stocks in Asia is the Bombay Stock Exchange. Among the biggest stock exchanges is one like this with a market capitalization of $3,612,985 million as of June 2023. Promoting and fostering fair and moral trading practices in securities transactions, as well as discouraging unethical behavior, is one of BSE’s main goals. Trading has to be done through a broker or a brokerage agency which charges a stipulated fee. Nonetheless, certain preferred investors who engage in significant transactions are granted direct access. The T+2 rolling settlement transactions, in which each transaction is completed in less than two days. SEBI is in charge of overseeing the BSE. [4]. It continuously updates the rules for its smooth operation. Some advantages of listing in the BSE are hassle-free capital Generation, Legal Supervision, Timely Information Display Adequate Pricing Rules, and Collateral Guarantee. The main financial exchange in the nation is the National Stock Exchange. On the advice of the Pherwani Committee, a group of prominent financial institutions came together to establish it. It consists of a variety of shareholding holdings from both domestic and foreign investors [5]. It was established with specific objectives in mind. It aims to create a nationwide trading establishment for

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equities, debt, and hybrid instruments. It also aims to enable faster settlement cycles, book entry settlement systems, and fulfill the latest international norms of securities markets. It serves an entirely automated system called the National Exchange for Automated Trading or NEAT. Each order received by the NEAT is assigned a unique number. If a match is not found, it is instantly included in an order book, where the sequence of orders to be matched is established based on price-time priority. Some benefits of Listing with the NSE are Comprehensive Visibility, a Premier marketplace, faster transactions, and better trade statistics [6]. An index of the stock market is a collection of stocks that can be used to represent the whole market or a particular subset of it. The NIFTY market index was introduced by the NSE. The NSE's flagship index, the NIFTY 50, is benchmark-based. Out of 1600 stocks, it displays the top 50 equity stocks that are traded on the stock exchange. The twelve Indian economic sectors that are represented by the NIFTY stocks are information technology, financial services, consumer goods, media and entertainment, metals, pharmaceuticals, telecommunications, cement and related products, automobiles, energy, fertilizers and pesticides, and other services. NIFTY is a component of the F&O and consists of several indexes, such as NIFTY 50, NIFTY IT, NIFTY Bank, and NIFTY Next 50. F&O [7] is a division of the NSE that oversees derivatives. The methodology used to calculate the NIFTY 50 indices is float-adjusted and market capitalization-weighted. Using this method, the level of the index displays the total market value of the constituent stocks over a specified base period. Broad indexes such as Sensex and NIFTY are used by mutual funds as a benchmark to evaluate their performance. The former offers a far more comprehensive benchmark of the Indian financial market because it is the larger of the two indexes. The purpose of this paper is to forecast the price of the stocks in the near future by performing a thorough time series analysis using the NIFTY 50 benchmark. Modern machine learning models are what we hope to use to complete this task [8].

II. Literature Review

Financial research has focused on predicting stock prices, with numerous approaches being investigated to improve prediction accuracy. Elliott and Siu (2005) established the groundwork by demonstrating the potential of machine learning methods for modeling complicated financial data [9], including neural networks, support vector machines, and genetic algorithms. Incorporating sentiment analysis, statistical models, and machine learning techniques, Yadav et al.'s (2016) thorough assessment presents a holistic view of stock market forecasting and gives readers a thorough knowledge of the field's current state. Nagpal and Jindal (2017) highlight the critical significance of news sentiment in predicting stock price changes by combining sentiment analysis with machine learning, offering a worthwhile direction for further investigation.

Bossy and Kobti (2019) further highlighted the significance of historical data and the use of recurrent neural networks (RNNs) [10] for time series forecasting, particularly in the prediction of stock price. Ingeniously combining news sentiment analysis with RNNs, Bhattacharjee et al. (2018) highlighted the importance of textual data in improving prediction accuracy. Bhatia et al.'s (2020) in-depth examination of deep learning methods including convolutional neural networks (CNNs) and long short-term memory (LSTM) networks shows their efficiency in handling complex financial data.

In addition to examining the use of deep learning techniques, Zhao et al.'s (2020) assessment of improving stock price prediction with deep learning [11] also discusses the field's practical difficulties and potential. In conclusion, Kakkar et al.'s (2018) review provides a comprehensive overview of machine learning algorithms, outlining their merits and limitations, and offering insightful information on the state of stock price prediction today. Collectively, these works highlight the value of sentiment analysis, time series forecasting [12], machine learning, and deep learning techniques in stock price prediction. They offer a solid framework for future research in this area, especially when viewed in the context of the NIFTY 50 dataset.

III. Exploratory Data Exploration

The weighted average of the 50 biggest Indian companies listed on the National Stock Exchange is used to calculate the benchmark NIFTY 50 index for the Indian stock market. This index and the BSE SENSEX are the two primary stock indices used in India. NSE Indices, formerly known as India Index Services & Products Limited, is the owner and manager of the Nifty 50 index. The index provides investment managers with exposure to the Indian market within a single portfolio with regard to 13 sectors of the Indian economy. Financial services are weighted at 39.47% in the NIFTY 50 Index. The next highest weighted categories are energy (15.31%), IT (13.11%), consumer goods (12.38%), automobiles (6.11%), and agriculture (0%).

The dataset used in this work is created from the official NSE India website.It has the following columns :

1) Stocks names of nifty 50
2) Open price for the current day
3) Price made high for the current day
4) Price made low for the current day
5) Change in rupees
6) Percentage of change from the previous day close
7) Traded volume in lakhs
8) Turnover in Crores
9) Stock 52-week high price (52w H)
10) Stock 52-week low price (52w L)
11) Stock price percentage change from one year
12) Stock price percentage change from last 30 days.

Figure. 1 shows the correlation map between all the attributes of the dataset. Some important points about the dataset are:

1) The adjusted price on the ex-date for the rights, dividend, bonus, and face value split is used to compute the percentage change.
2) Bonus, split, and rights corporate actions are factored into prices at a 52-week high and 52-week low.

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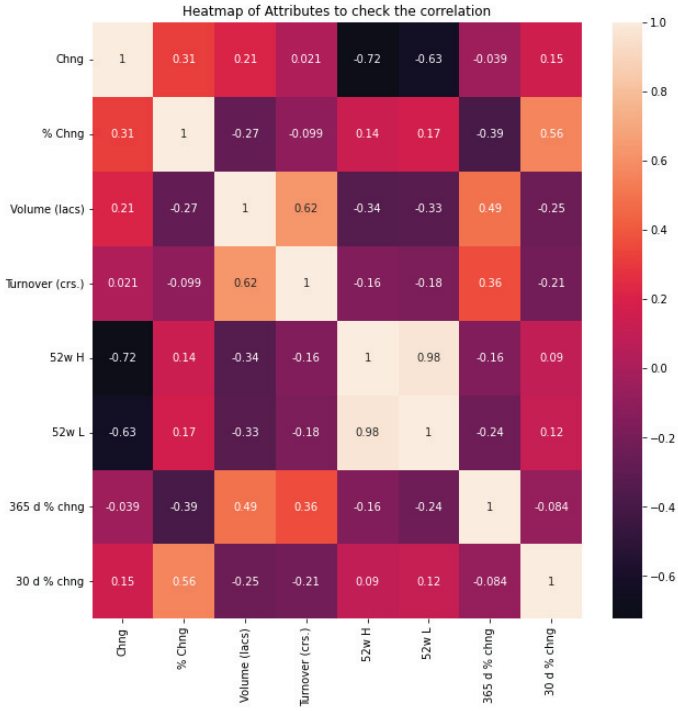

Fig. 1: Heatmap for showing a correlation between attributes in the dataset.

3) In line with business actions, values for the 30-day and 365-day percentage changes are adjusted.

A. Annual Growth Rate

The Annual Growth Rate, or AGR, is the annual growth of your investments over a specified period of time. Stated differently, it shows how much money you have earned each year from your investments over a given period of time. The percentage annual growth of the Nifty 50 stocks is plotted in Figure 2. As can be seen, Tata Motors provided the highest annual returns—more than 167.95%. In contrast, the Hero Motor Co. stock had the lowest annual returns—less than -16.02%.

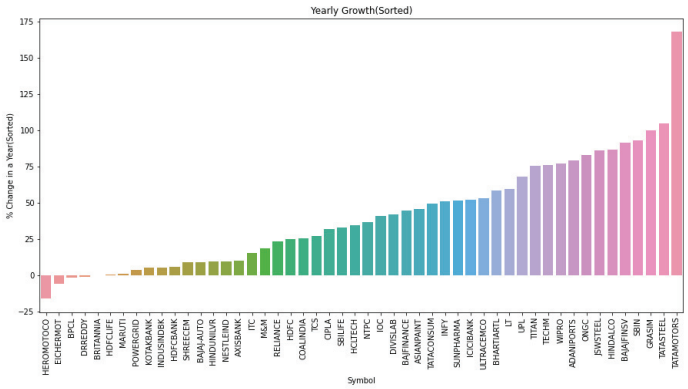

Fig. 2: The annual growth of Nifty 50 stocks.

B. Monthly and Yearly Returns

Monthly returns are term returns that have been rescaled to one month. Investors can now assess the returns on different assets they have owned for different lengths of time thanks to this.
(Closing Price on Last Day of Month / Closing Price on Last Day of Previous Month) - 1 is the monthly return.
The bulk of the companies have demonstrated negative returns if an investor stays invested in them for a month, as can be seen in Figure 3, a bar plot for monthly returns. Therefore, the investment period ought to extend beyond a month.

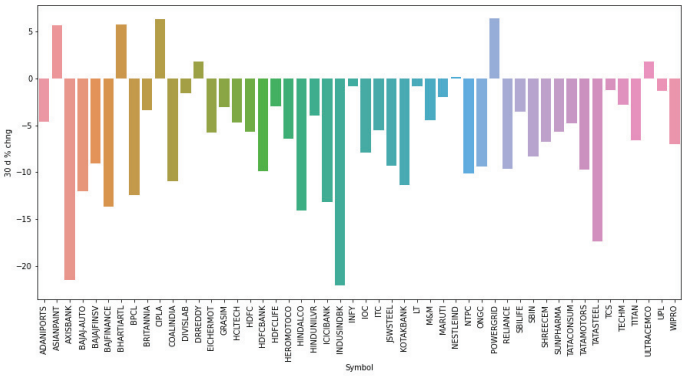

Fig. 3: Monthly Returns of Stocks.

The profit or loss on an investment over a full year is called the Yearly Return. The yearly returns are expressed by multiplying the rate of return by 12 if it is determined on a monthly basis. The yearly return bar plot presented in Figure 4 indicates that most companies have demonstrated at least a 10% return on investment over a one-year period.

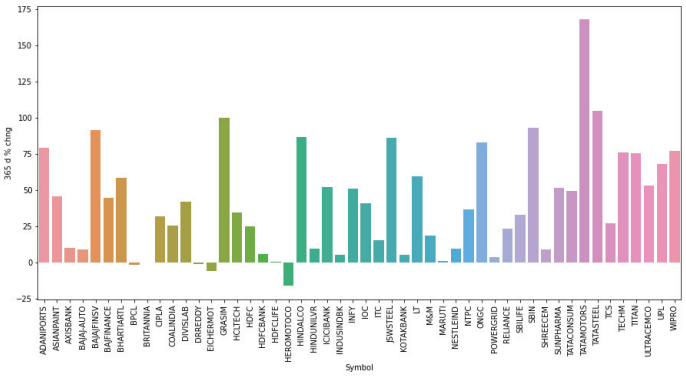

Fig. 4: Yearly Returns of Stocks.

C. Opening and closing prices

When an exchange begins for business each day, a securities trades at its starting price. An opening price is not identical to the previous day's closing price. The final price a share of that stock was purchased for during regular business hours on the

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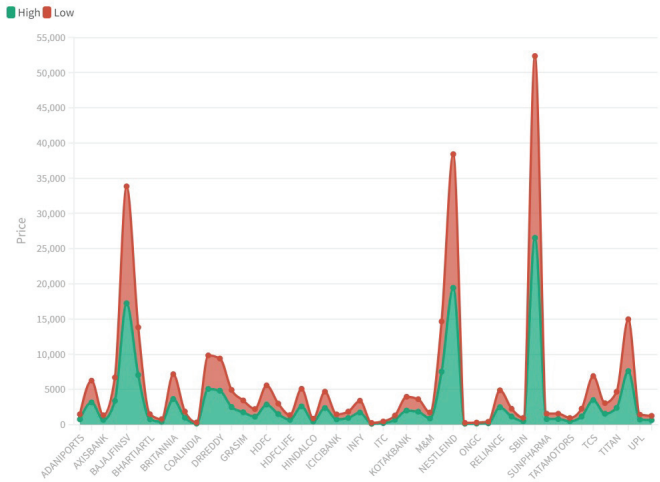

Fig. 5: The opening price of stocks.

exchange where it is traded is the listed closing price. From Figure. 5 we find that the most significant opening was by: SHREECEM (opening price at Rs. 26,450), NESTLEIND (at Rs. 19,148.85) and BAJAJFINSV (at Rs. 17,200).

D. Banking Sector

Investors can use the Nifty index as a benchmark to see how Indian bank companies have performed on the capital market. There are twelve banking stocks in the index. BajajFinsv, BajajFinance, IciciBank, and SBIN outperformed in the Banking sector, giving more than 40% returns in a Year. AxisBank and IndusInd Bank are the worst performers with more than 20% loss of investors in a single month.

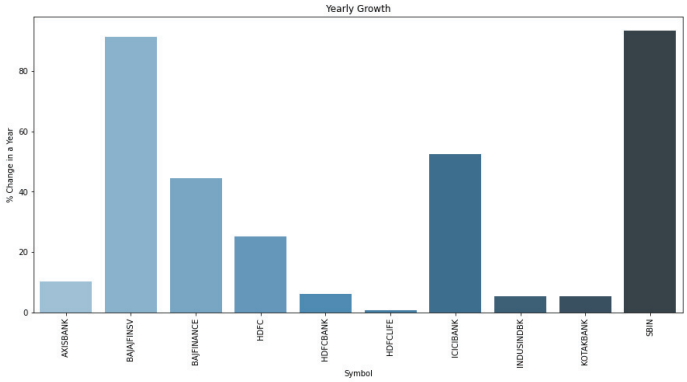

Fig. 6: Best performing banking sector stocks.

IV. Methodology

A. Data Pre-Processing

The most important first step in our study is data pre-treatment, which lays the groundwork for deft inferences.The purpose of this crucial stage is to prepare the raw data over careful examination by cleaning, transforming, and organizing it. Improving data quality is the primary focus of data pre-processing, which prepares the data for in-depth analysis and

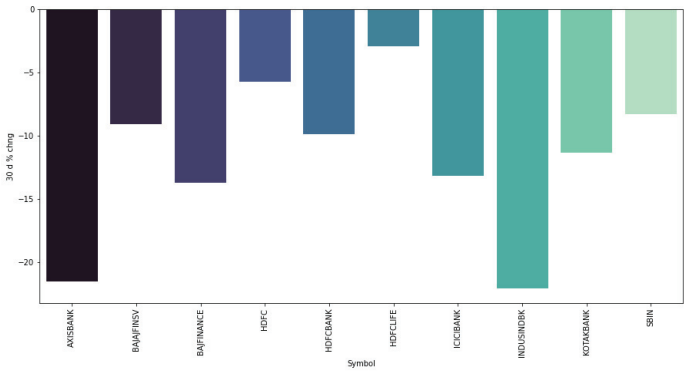

Fig. 7: Worst performing banking sector stocks

interpretation. Within the subsequent parts, we describe the painstaking measures we took to clean up and improve our dataset so that we could draw conclusions from the complicated and frequently noisy raw data. Using data processing techniques ensures the accuracy of data and readability. This entails managing different data kinds, guaranteeing consistency, and structuring the dataset.

In order to prepare the financial dataset for comprehensive analysis, we have meticulously preprocessed it using stock prices from the NIFTY-50 index. The preprocessing pipeline consists of several vital steps.

1) Data Import
2)Data Cleaning and Transformation
3)Data Slicing
4)Data Visualization
5)Time Series Decomposition

We carried out the required data transformation and cleaning using the 'pandas' package. This involved transforming the 'Date' column into a 'datetime' format and designating it as the index in order to guarantee temporal data consistency. Following that, we illustrated the usefulness of data slicing by allowing the selection of particular date ranges, which allows for targeted study on chosen time periods. We used data visualization to have a better understanding of the temporal patterns and trends in the dataset. Time series plots and kernel density estimation plots were created using the 'matplotlib' and 'seaborn' libraries. In addition, we used the 'statsmodels' library to perform time series decomposition, which helped us uncover hidden patterns in the dataset by breaking it down into its trend, seasonal, and residual components.

B. Transfer learning with pre-trained Convolutional Neural Networks

1) ARIMA: ARIMA or Auto-Regressive Integrated Moving Average is a popular method for Time-series forecasting tasks such as the one we are undertaking in this paper [13]. It comprises 3 parts: AR is the auto-regressive component. It simulates how the current data point and its historical values relate. It makes the assumption that a time series' current value can be predicted by combining its prior values in a linear fashion. The Integration (I) component deals with differencing

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the time series data to make it stationary [14]. Stationarity is crucial in time series analysis because it simplifies the modeling process. The relationship between the current value and previous forecast errors is taken into consideration by the MA component. The current value is assumed to be determined by a linear combination of previous forecast errors. In our work, we Implement an ARIMA model for the prediction that utilizes automatic order selection. For this, we use the auto_arima function in the pmdarima module [15].

2) SARIMAX: SARIMAX is an enhancement or expansion of the conventional ARIMA model, adding seasonality as an additional variable to the time series forecasting procedure [16]. SARIMAX allows us to model the impact of external variables (exogenous variables) on the time series data. However, SARIMAX also considers seasonal patterns, which can be crucial for forecasting when data exhibits recurring trends [17] over specific time intervals, such as daily, weekly, or monthly seasons. SARIMAX is particularly valuable when dealing with time series data that exhibits both seasonality and dependencies on external factors like the stock market [18] [19] [20].

C. LSTM

An LSTM (long short-term memory) architecture with multiple layers is used in this study. To overcome the vanishing gradient problem, LSTMs—a particular type of RNN—were developed, allowing them to effectively capture long-range dependencies [21]. This architecture is critical for capturing and learning complex patterns, trends, and seasonality in time series data for stock prices.

To analyze stock market data, the methodology involves creating a Long Short-Term Memory (LSTM) model with an emphasis on the "BHARTIARTL" stock [22]. Recurrent neural network (RNN) architectures are often used for sequential data analysis, and one such architecture is the LSTM. Because it can recognize temporal dependencies and patterns, LSTM can be a useful tool for modeling and predicting stock prices in the context of stock market data [23].

The TensorFlow (tf) library is used to construct the LSTM model. Since layers are added one after the other, the model is sequential. An LSTM layer with 32 units makes up the first layer. It uses the training data's shape as input, with "trainX" standing in for the input data. The last LSTM layer is indicated by the "return_sequences" parameter being set to "False," meaning that sequences will not be returned. A Dense layer with one unit is added after the LSTM layer. The predicted stock price is the output that this layer is in charge of producing.

The Adam optimizer and the mean squared error as the loss function are used to compile the model. For regression issues like stock price prediction, mean squared error is frequently used, and the Adam optimizer is a powerful optimization technique [24].

D. XGBoost

In the field of time-series forecasting, the XGBoost (Extreme Gradient Boosting) architecture is a potent and well-known machine learning algorithm, particularly for forecasting stock prices [25]. Its significance can be found in several important ways: As an ensemble learning technique (typically decision trees), XGBoost combines predictions from a number of weaker models to produce a stronger and more precise predictive model. The complex relationships and patterns in the stock price data are better captured by the ensemble approach than by individual models alone [26]. Gradient Boosting: The XGBoost algorithm reduces residuals, or the differences between expected and actual values, from prior iterations in order to iteratively improve the model. XGBoost is effective at lowering errors and increasing prediction accuracy because of its iterative learning process. The objective function in XGBoost is typically written as:

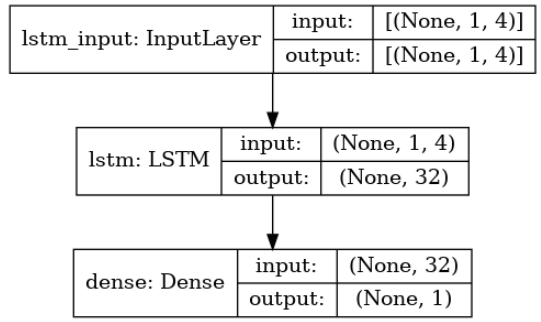

Fig. 8: Architecture Of LSTM

$$Obj(\theta) = L(\theta) + \Omega(f) \tag{1}$$

Where:

$Obj(\theta)$ - Objective function
$L(\theta)$ - Loss function
$\Omega(f)$ - Regularization term

The boosting process in XGBoost involves iterative updates to the predictions. The predicted value for a given data point x_i after t boosting rounds is given by:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \gamma_t h_t(x_i) \tag{2}$$

Where:

$\hat{y}_i^{(t)}$ - Prediction after round t
$\hat{y}_i^{(t-1)}$ - Prediction from the previous round
γ_t - Learning rate for round t
$h_t(x_i)$ - Prediction of the newly added tree

XGBoost is more considered on the tooling of the time series data due to its ensemble learning, gradient boosting & regularization techniques [27] [28].

V. Result Analysis And Inference

The ARIMA model was trained on the data that spans till 01-01-2020 and the rest of the data we used for testing. After compiling and testing our model we can see the predicted trend of the stock market along with the actual values. Figure.

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10 shows that the model performed very well in predicting the VWAP of the stocks with very little deviation from the original values.

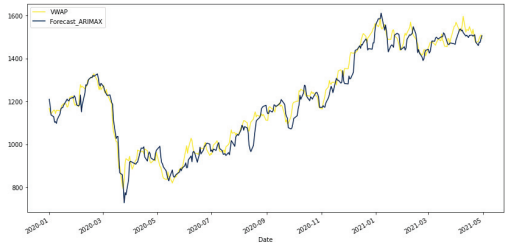

Fig. 9: Prediction of VWA price by the ARIMA model

The SARIMAX performed the task of predicting or forecasting the trades of particular stocks. For example, for the stock labelled ”TCS”, ”Wipro” and ”HCL”.

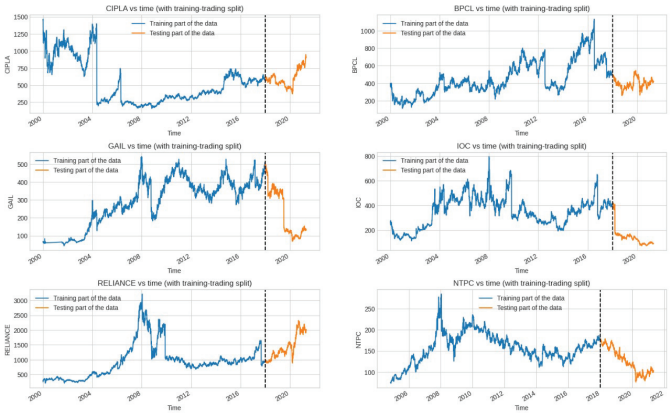

Fig. 10: Training and Testing Data Split Visualization for Stock Price Prediction using SARIMAX.

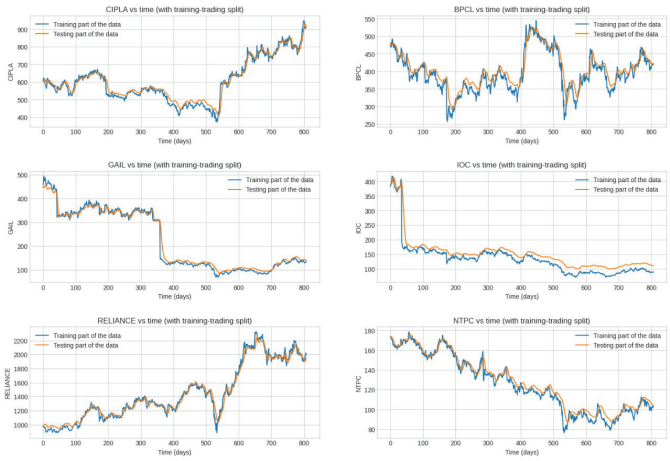

Fig. 11: Stock Price Predictions vs. Actual Values Comparison using LSTM.

The LSTM model, trained on historical stock data, demonstrates its effectiveness in predicting stock prices. It leverages its ability to capture temporal dependencies and sequences, making it well-suited for stock market data analysis. The model’s output provides valuable insights into the future trends of the ’BHARTIARTL’ stock. Through the use of the mean

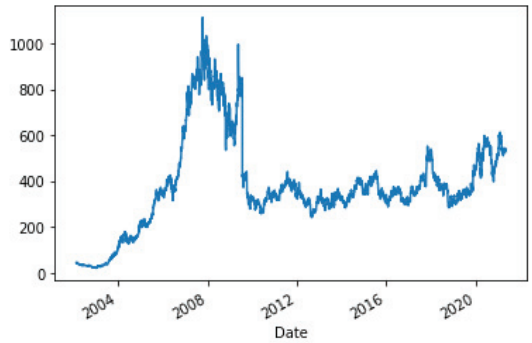

Fig. 12: Volume-weighted average price of Barati Airtel for LSTM model.

squared error loss function, the LSTM model offers accurate predictions, aiding in informed investment decisions. This underscores the significance of LSTM architectures in stock market forecasting. The LSTM model is evaluated based on the ”Mean Squared Error” & ”Root Mean Squared Error” valued at 24.4 & 4.9

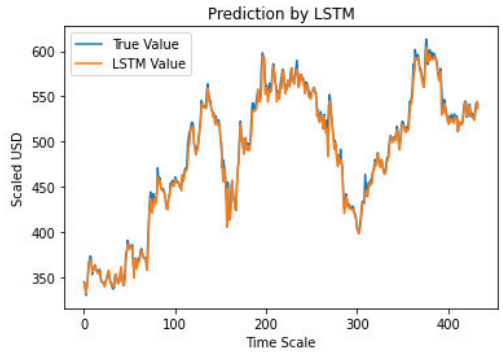

Fig. 13: True Vs Predicted values of BaratiAirtel using LSTM

In the final phase of our analysis, we implemented TimeSeriesSplit with a five-fold approach to assess the performance of the XGBoost model in our time-series prediction. The graphical representation of the cross-validation results provides a clear visualization of the model’s performance across different time periods.

The x-axis in each subplot signifies time, underscoring the fundamental role of time in time-series data. The ’Training Set’ and ’Test Set’ lines on each subplot represent the respective portions of the dataset utilized for training and testing during each fold. Notably, a vertical dashed line serves as a visual indicator, highlighting the boundaries of the test set for each fold.

This meticulous time-series cross-validation methodology serves a dual purpose. First, it offers a comprehensive evaluation of the model’s effectiveness over diverse time periods, al-

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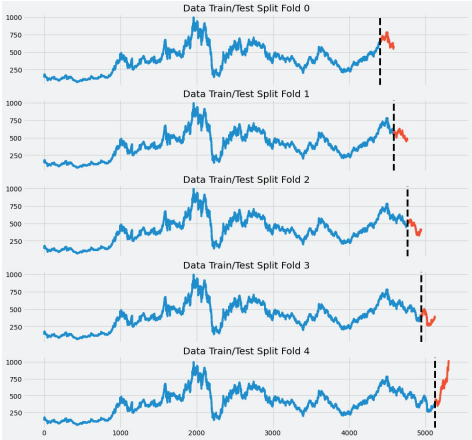

Fig. 14: Prediction with XGBoost

lowing for a robust assessment of its generalizability. Second, it underscores the critical importance of accommodating temporal dependencies in the analytical process.By considering the temporal aspects inherent in time-series data, this approach contributes to a more accurate and reliable assessment of the predictive model's capabilities.

VI. Conclusion and Future Scopes

In this comprehensive study, our primary objective was to conduct an in-depth exploration and application of various machine learning models for stock price forecasting. The models examined include ARIMA, SARIMAX, RNN, LSTM, and XGBoost, each contributing unique insights into the challenges and opportunities in predicting stock prices.

Our investigation yielded valuable insights into the strengths and weaknesses of each model, emphasizing the critical role of factors such as hyperparameter tuning, feature engineering, and the quality of input data in ensuring the success of machine learning models in stock price prediction. It is essential to acknowledge that no single model is without limitations, and a nuanced understanding of these constraints is crucial for effective implementation.

Recognizing the inherent unpredictability of stock price movements due to external influences, such as news headlines, market sentiment, and economic indicators, we emphasize the importance of acknowledging and accounting for these uncertainties in any predictive modeling approach.

Looking forward, numerous avenues for further exploration and advancement in this domain present themselves. Hybrid models, leveraging the strengths of diverse methodologies, offer a promising direction. Improved feature engineering, incorporating alternative data sources beyond traditional financial indicators, is an area ripe for development. Enhancing model interpretability through Explainable AI can contribute to better understanding and trust in the predictions.

Additionally, the application of reinforcement learning for optimizing trading strategies, transfer learning to leverage knowledge from related domains, and delving into high-frequency trading strategies are avenues that warrant further investigation. The integration of sentiment analysis and real-time data processing can also enhance the robustness of predictive models. Ethical considerations and robust risk management practices remain integral aspects of advancing machine learning applications in the financial domain.

In conclusion, this study not only sheds light on the current landscape of stock price forecasting using machine learning models but also outlines promising directions for future research and development in this dynamic and challenging field.

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